

Global Facilities - Design & Construction Standard - General - Base Build vs Tool Install Scope

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[ECN History](#)[Submit a Change Request](#)

1.0 Purpose

- 1.1 This document provides the overview of Global scope of work overview, definition and boundaries that shall be followed during the new Greenfield Construction, Yellow field Expansion or major remodel projects.
- 1.2 The document is intended to provide a guideline of proper scope delineation between Base Build and Tool Installation, roles and responsibility. Refer to the individual System BKM Documents or Global System Standards for the exact system specification.
- 1.3 Providing alignment across Facilities sites:
 - 1.3.1 The main objective would be to protect our internal customers and eliminate recurrence of events using globally aligned methodologies to achieve Best-In-Class Facilities performance.
 - 1.3.2 Sites are required to review the guidelines provided in this global standard specification document & align their procedures or scope delineation to ensure compliance where applicable.

2.0 Scope

- 2.1 This document applies to all Facilities systems outlined in the document.
- 2.2 Site(s) and Person(s) impacted
 - 2.2.1 This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites, including Frontend Manufacturing (FE) and Backend Assembly & Test (AT) sites.

3.0 Definitions and Acronyms

Term/Acronym	Definition / Description
ACM	Air Composition Monitor
AHJ	Authority Having Jurisdiction
AMHS	Automated Material Handling System
BB	Base Build
CCDP	Chemcassette Detection Panel
CDA	Clean Dry Air
CDU	Chemical Dispense Unit
CM4P	Four-Point Continuous Monitor
DB	Distribution Board
DI	De-Ionized
DVB	Distribution Valve Box
FAB	Semiconductor fabrication facility
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs
FM or FMG	Factory Mutual Global Insurance Company
FOP	Feed Out Panel
FOUP	Front Opening Unified/Universal Pod
FTIR	Fourier Transform Infrared Spectroscopy
GC	Gas Cabinet
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team
HPM	Hazardous Production Materials
HVM	High Volume Manufacturing facility
LDC	Liquid Dispense Cabinet

Term/Acronym	Definition / Description
LOTO	Lock-Out, Tag-Out
LPOC	Lateral Point-of-Connection
LSS	Life Safety System
MEP	Mechanical, Electrical & Process
MT	Main Tool
NG	Natural Gas
PCW	Process Cooling Water
PDP	Power Distribution Panel
PFA	Perfluoroalkoxy
PIU	Plug In Unit
POU	Point-of-Utilities
PP	Power Panel
PV	Process Vacuum
RIO	Remote Input/Output
RO	Reverse Osmosis
SCADA	Supervisory Control and Data Acquisition
SS	Stainless Steel
TGM	Toxic Gas Management
TI	Tool Install
TPOC	Tool Point-of-Connection
UPW	Ultra-Pure Water
VMC	Valve Manifold Box

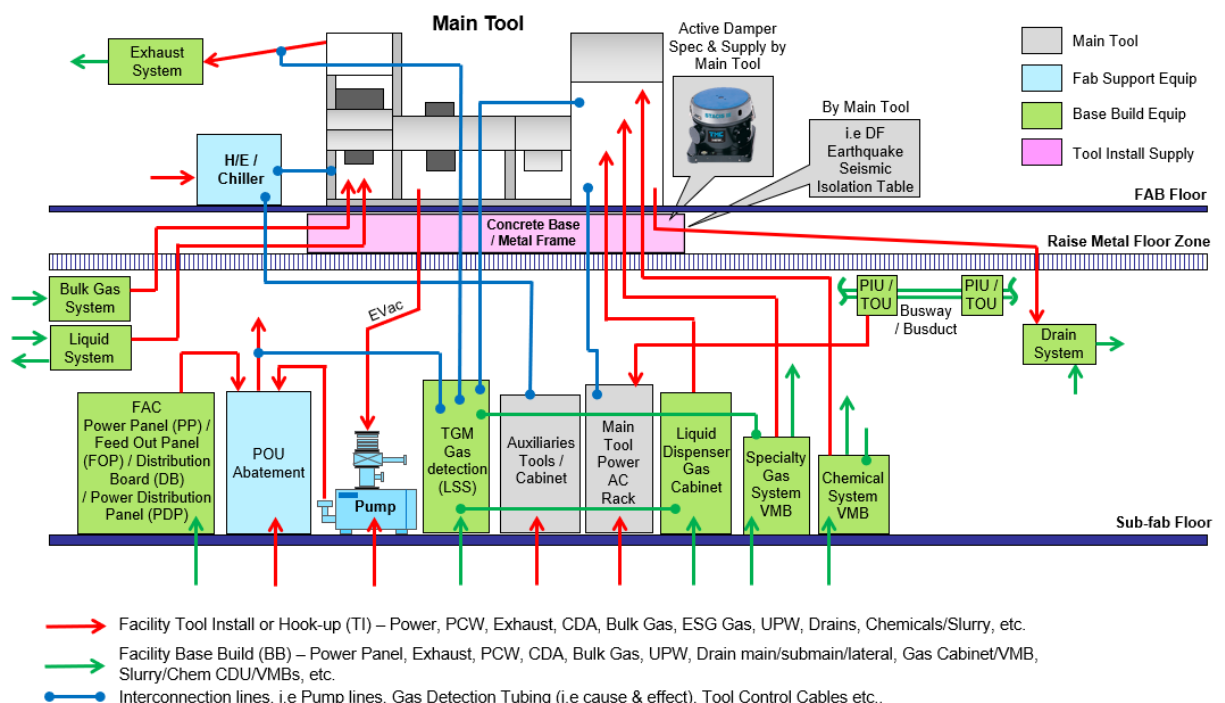
4.0 Roles and Responsibilities

Roles	Responsibilities
GFTT	• Develop and maintain document (Base Build aspect) • Ensure Best Known Methods (BKM), lessons learned, are documented in document • Perform periodic document review and update document • Review and update scope delineation of new system when introduce and recommend new Facilities system for both new Greenfield FAB Construction projects, and for existing Site upgrade projects.
FCT	• Review and provide technical feedback on document • Support the Site teams by providing knowledge and technical expertise • Share new lessons learned to GFTT, to determine if they are BKMs that can be captured in future revision of document • Review and update scope delineation of new system when introduce and recommend new Facilities system for existing Site upgrade projects.
GFTI	• Develop and maintain document (Tool Install aspect) • Ensure Best Known Methods (BKM), lessons learned, are documented in document • Perform periodic document review and update document • Review and update scope delineation of new utilities when new utilities requirement from Tool for both new Greenfield FAB Construction projects, and for existing Site upgrade projects.
Site Project Construction Teams	• Review and provide technical feedback on System Standard • Incorporate System Standard into Project Engineering, Design, and Construction • Document project deviations, exceptions, and exclusions from System Standard • Share new lessons learned to GFTT lead & GFTI lead, to determine if it is Best Known Method (BKM) that can be captured in future revision of System Standard
Site Facilities Teams	• Use System Standard as technical reference • Share new lessons learned to GFTT lead, to determine if it is Best Known Method (BKM) that can be captured in future revision of System Standard
Site Tool Install Team	• Use System Standard as technical reference • Share new lessons learned to GFTI lead, to determine if it is Best Known Method (BKM) that can be captured in future revision of System Standard

5.0 Scope of Work Definition

- 5.1 Base Build:** Base Build normally referred to a few elements such as, primary structure, building envelope, MEP & distribution system. However, in this context, Base Build are referring to upstream MEP system from main plant room until facilities base build isolation point. Facilities base build isolation point inclusive but not limited to Lateral Point of Connection (LPOC); or valve connection points (i.e GC/LDC/VMB); or any Point of Utilities (POU); or outgoing cabling work from circuit breaker MCCB or Plug-in-Unit / Tap-off-Unit TOU.
- 5.2 Tool Install:** Tool Install refer to secondary distribution lines from main system utilities. Tool Install scope is from Lateral Point of Connection (LPOC); or valve connection points (i.e GC/LDC/VMB); or any Point of Utilities (POU); or outgoing cabling work from circuit breaker MCCB or Plug-in-Unit / Tap-off-Unit TOU from the upstream Facilities system isolation point(s) onward to Tool Point of Connection (TPOC). For more scope of work & scope boundaries, please refer to respective Tool Install trades standard specification documents.
- 5.3** Commonly the distribution lines referred as piping (i.e liquid), tubing (i.e gas), power cables, chemical, drains, equipment vacuums lines etc. Base Build and Tool Install scope of work overview illustrated in **Figure 1**.

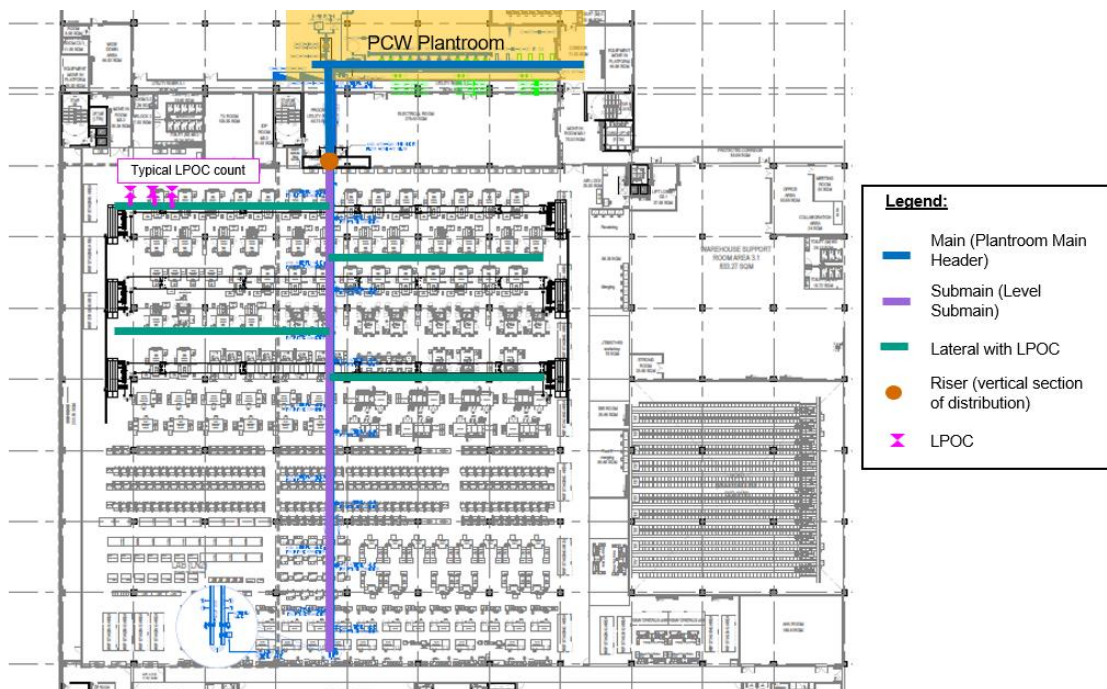
Figure 1 – Overview of Scope Ownership & Boundaries



6.0 Scope Boundaries

6.1 Base Build Scope Boundary

- 6.1.1 Base build scope are the main utilities plant and primary distribution system that cover the full area of production or specific are of production with special need. Piping carrying foam solution shall be sized as if carrying plain water.
- 6.1.2 The base build distribution system shall be completed with an isolation point or bypass module (RIO) or any mean of isolation, to comply with CoHE or any forms of hazardous energy isolation & safety interlock capabilities.
- 6.1.3 For any tool location/LID/HEX resulted to alteration & addition (A&A) of building infrastructure, additional LPOC hot-tapped, or any extension of lateral, or any new BSGS or Chemical delivery system, even any other Major System Capacity expansion ancillaries, it would be fall under cost and scope of Base Build.
- 6.1.4 Whereas any tool position, which resulted to any existing equipment relocation, or alteration, or any modification work to be carried out, or any emergencies equipment (Hose reel, Fire-alarm, Emergency Rain Shower etc) relocation, it shall be part of site Manufacturing Expense, based on Finance & Accounting Rules and definition.
- 6.1.5 Any other ancillaries which not mentioned in this scope delineation, required Base Build system owner, as well as requester to further clarify the scope of work, and cost ownership.
- 6.1.6 Base Build Distribution System (lateral base utilities) definition are as below:
 - a. **Main** – Large piping/ducting main header at plantroom which connect main equipment or main piping/ducting connecting each building.
 - b. **Submain** – A branch from main header piping/ducting and supply to different level or compartment.
 - c. **Lateral** – A branch from the submain or main piping/ducting with multiple LPOC that install according to downstream user bay or zone.
 - d. **LPOC** – Lateral Point-of-Connection served as isolation point or end of Base Build Distribution System
- 6.1.7 Refer to **Figure 2** for illustration of base build distribution system (lateral base utilities).

Figure 2 – Example Illustration of Base Build Distribution System (Lateral Base Utilities)

6.1.8 See **Table 1** for details of BB isolation point and other ancillaries that under BB scope.

Table 1 – Base Build isolation point, scope & other ancillaries for various utilities.

System	BB Isolation Point	BB scope & other ancillaries
Bulk / Process Gas (Lateral)	Lateral LPOC	Lateral expansion valve, LPOC
Specialty Gas	Valve/Isolation Point, terminal block for communication interface, LSS shutdown and interlock	LDC, VMB, DVB, GC
Exhaust	Damper c/w blind plate	Exhaust Hot Tap

System	BB Isolation Point	BB scope & other ancillaries
Electrical	System c/w MCCB, Isolator, Breaker	Tap Off Unit, Plug In Unit, DB, PP, FOP
TGM / LSS	RIO Bypass Module, tubing plug.	FTIR (ACM), CCDP (Vertex), CM4P, PDP, RIO
Remote Monitoring System	RIO Bypass Module, tubing plug.	RIO, SCADA (Wonderware)
Waste Chemical / Wastewater Drain (Lateral)	Lateral LPOC	Lateral expansion valve, LPOC
Process Chemical (VMB)	Valve/Isolation Point	VMB
Liquid (PCW/NPW/UPW/RO/DI)	Lateral LPOC	Lateral expansion valve, LPOC

6.1.9 Site-specific differences based on the legacy buildings or systems, formal review and approval is required by GFTT, GFTI, FCT, OCT, Planning team.

6.2 Tool Install Scope Boundary

- 6.2.1 Tool install scope are connecting utilities from base build isolation point to Tool Point of Connection (TPOC)
- 6.2.2 The tool install/ hookup scope, it should be deemed as a transitional point from mainstream System, a.k.a BB (Upstream/Source) to Hookup (Downstream).
- 6.2.3 The tool install/ hookup scope, it should be deemed as a transitional point from mainstream System, a.k.a BB (Upstream/Source) to Hookup (Downstream).
- 6.2.4 See **Table 2** for details of TI/Hookup scope to Main Tool (MT).

Table 2 – Tool Install / Hookup scope to main tool

System	Hookup Scope to Main Tool (MT)
Bulk / Process Gas (Lateral)	SS Tubing from Lateral to MT
Specialty Gas	SS Tubing from Equipment to MT
Exhaust	Duct from Lateral to MT
Electrical	Cable from Panel to MT
TGM / LSS	PFA Tubing from Equipment to MT
Remote Monitoring System	Control cable from Equipment to Panel
Waste Chemical / Wastewater Drain (Lateral)	Hookup Line from Lateral to MT
Process Chemical (VMB)	Hookup Line from VMB to MT
Liquid (PCW/NPW/UPW/RO/DI)	Hookup Line from Lateral to MT

7.0 Document Control

7.1 **Approval:** GLOBAL_FAC_GFTT_DOC_APPR

7.2 Notification

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_FAC_NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY

7.3 Final Viewer Access

All Micron Team Members at all Sites

7.4 Retention

Adherence to the requirements specified in
[Global Facilities - Document Control Procedures](#)

7.5 Review

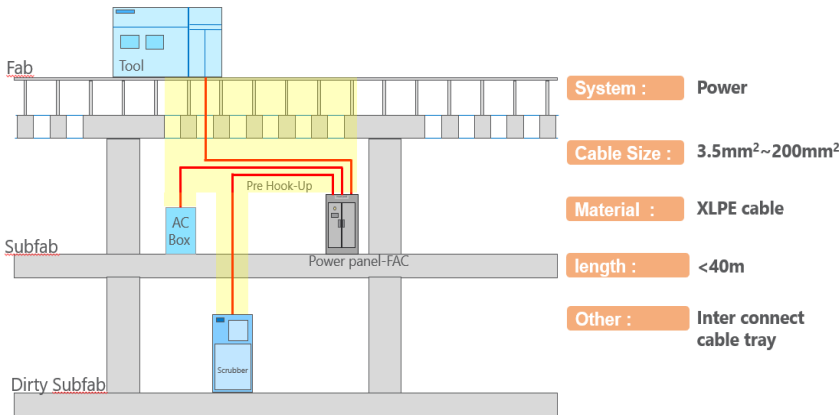
Biennially (two years)

8.0 Revision History

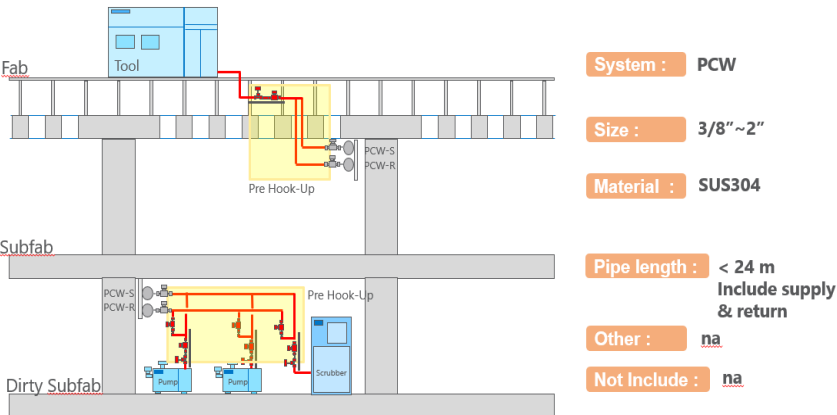
Revision #	Section Changed	Details of Changes	Date
0	NA	New document	04/14/22

Appendix A – Tool Install / Hookup Scope Illustration based on Various Utilities

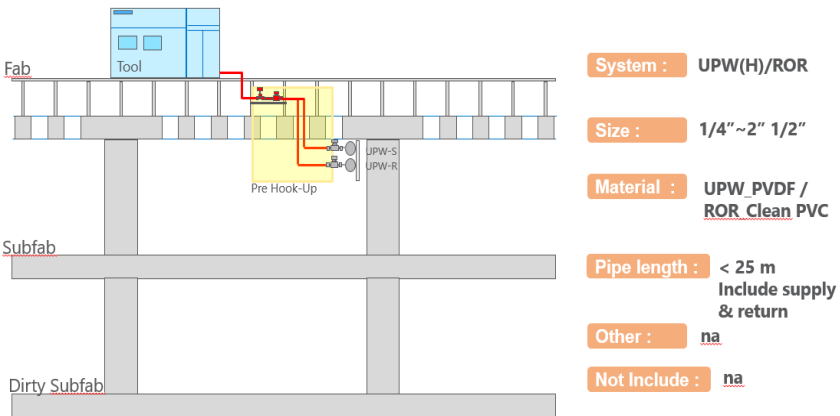
Hookup SOW - Power



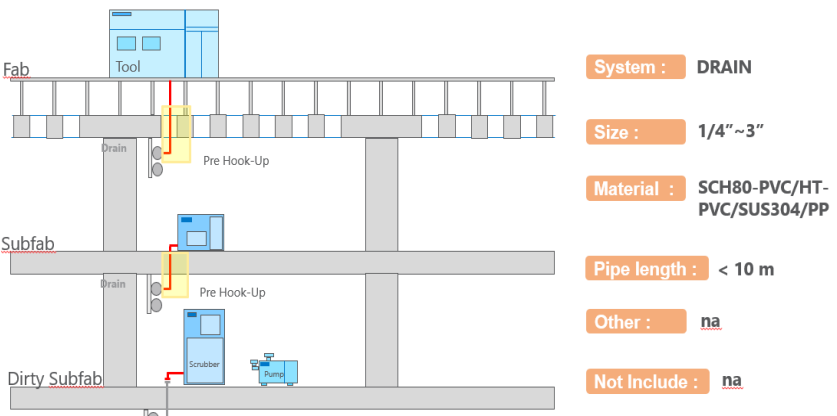
Hookup SOW - PCW



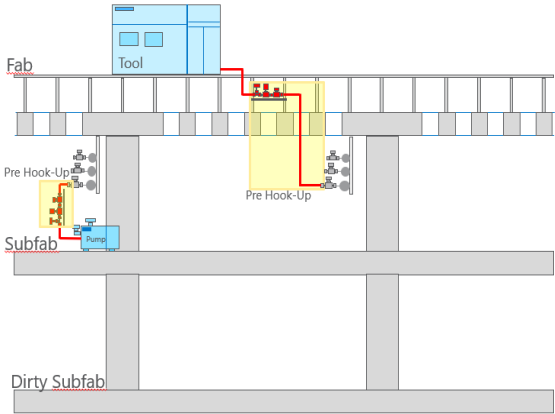
Hookup SOW - UPW(H)/ROR



Hookup SOW - Drain

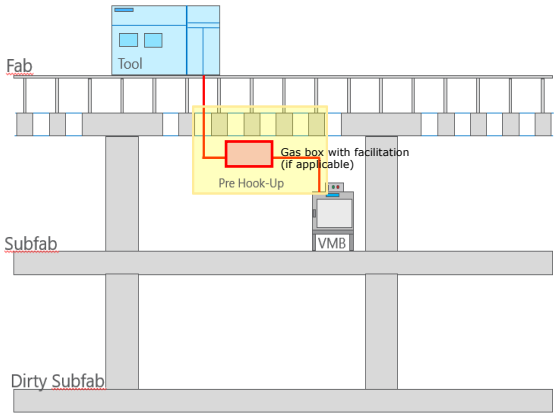


Hookup SOW - Bulk Gas



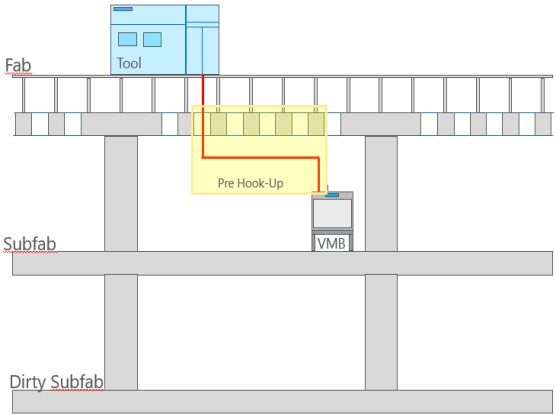
- System : Bulk Gas
- Size : 1 / 4" ~ 40A
- Material : SUS316L EP
SUS316L BA
- Pipe length : < 24 m
- Other : na
- Not Include : na

Hookup SOW - Specialty Gas



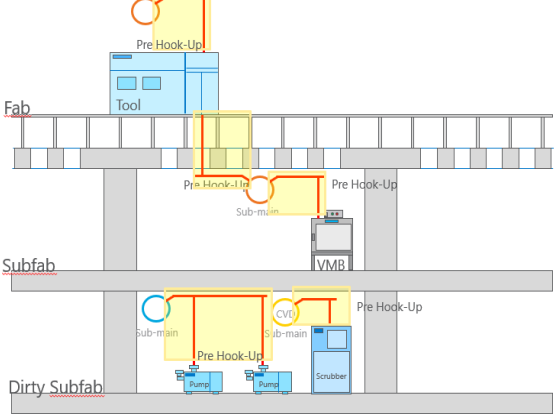
- System : Specialty Gas
- Size : 1 / 4"
1 / 4" * 1 / 2"
- Material : SUS316L EP-
4M / Coil Tube
- Pipe length : < 50 m
- Other : Heating
Double Tube
- Not Include : na

Hookup SOW - Chemical



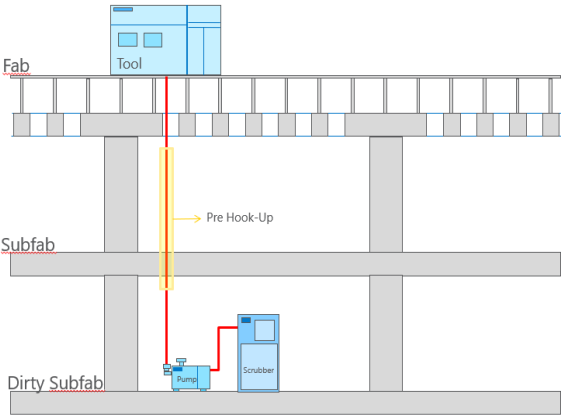
- System : Chemical
- Size : 3 / 8" ~ 1"
- Material : PFA + CPVC /
SUS316L EP
- Pipe length : < 30 m
- Other : na
- Not Include : na

Hookup SOW - Exhaust



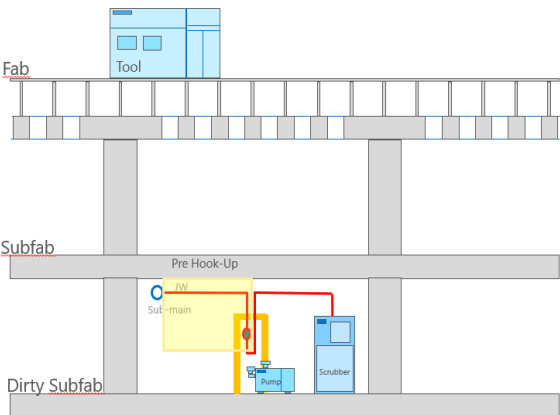
- System : Exhaust
- Size : 4" ~ 12"
- Material : Galvanized
SUS304
Coating ETFE
- Pipe length : < 20 m
- Other : na
- Not Include : na

Hookup SOW - Pumping line



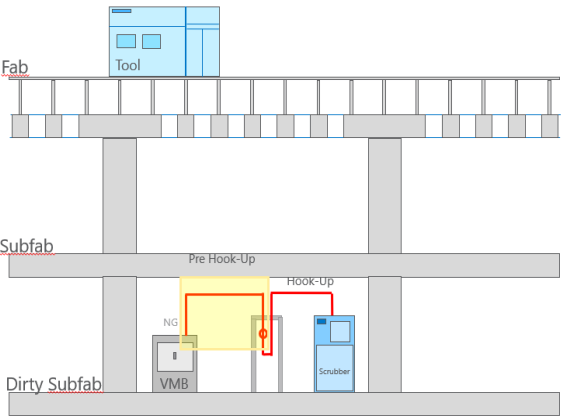
- System : Pumping line
- Size : 2"~6"
- Material : SUS304
- Pipe length : < 20 m
- Other :

Hookup SOW Work – NPW



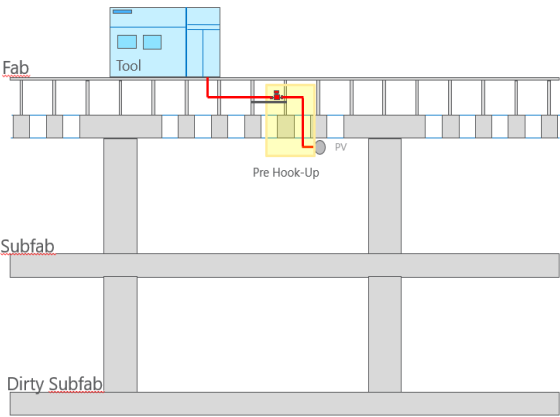
- System : JW
- Size : 3/8"~1"
- Material : SUS304
- Pipe length : < 12 m
- Other : na
- Not Include : na

Hookup SOW - NG



- System : NG
- Size : 3/8"~3/4"
- Material : SUS304
- Pipe length : < 45 m
- Other : na
- Not Include : na

Hookup SOW – Process Vacuum



- System : PV
- Size : 3/8"~3/4"
- Material : GIP
- Pipe length : < 12 m
- Other : na
- Not Include : na